Bay Area Real Estate Prediction Model and Dashboard

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1. Introduction

The San Francisco Bay Area housing market is one of the most expensive and complex in the country. Median sale prices across Santa Clara, San Mateo, and Alameda counties range from $1.15M to $1.50M, with prices varying more than 40% across zip codes within the same county. Yet the tools available to buyers have not kept pace. A serious buyer today needs to cross-reference Zillow for listings, Redfin for price history, a mortgage calculator for affordability, and separate resources for neighborhood quality with no way to bring it all together. This project fills that gap with an end-to-end system that predicts Bay Area home prices, filters the market by personal affordability, and scores neighborhoods on two original metrics, all within a single interactive dashboard.

2. Problem Definition

Our goal is to answer the following question in a simple-to-use dashboard: Given a buyer's income, available savings, and neighborhood preferences, which Bay Area homes are realistically affordable, and what is a reasonable expected price for those homes?

Current platforms like Zillow and Redfin are built for browsing and require utilizing multiple tabs to make an educated decision. They do not connect price estimates to a buyer's actual budget, and they offer no way to filter neighborhoods by quantifiable quality signals like park access or walkability. Every buyer sees the same impersonalized results, irrespective of their priorities.

Our system addresses this by combining three features that no platform currently offers together: an Gradient Boost price prediction model trained on around 50,000 Bay Area transactions across 143 zip codes, an affordability filter that maps a buyer's income and down payment to a realistic price ceiling, and two novel neighborhood scores; a Kid-Friendly Score built from OpenStreetMap park data and a Cool Score built from walkability, restaurants, and local attractions. The result is a dashboard where a buyer inputs their finances and preferences to immediately see which neighborhoods they can afford and why.

3. Literature Survey

The machine learning literature on housing price prediction is mature and consistent: ensemble tree methods outperform linear hedonic models, and careful feature engineering often matters more than model choice. Park & Bae (2015) established this hierarchy on Korean transaction data; we replicate and extend their findings with Gradient Boosting in Bay Area records. Baldominos et al. (2018) benchmarked 12 algorithms and found Gradient Boosting achieves 5–8% MAPE on well-structured datasets which is consistent with our own 8–9% MAPE in the $500K–$2M price range. Varma et al. (2018) showed that location-based feature engineering (zip-level price averages, distance to amenities) drives as much improvement as switching model families, which directly motivated our target-encoded location features. Bency et al. (2017) pushed performance further with deep learning on satellite imagery, but at the cost of interpretability; we deliberately chose explainable ensembles. Adetunji et al. (2022) validated Random Forest on the Boston Housing dataset; we scale comparable methodology to 50,000+ records across three counties. Mora-García et al. (2022) flagged COVID-era distortions in Spanish housing data, informing our decision to filter training data to 2019 - 2024 rather than extending further back.

On the application side, Crompton (2001) meta-analyzed 30 studies and found that proximity to parks adds 10–20% to nearby property values, making the empirical basis for including our Kid-Friendly Score as a predictive feature. Boeing (2017) introduced OSMnx, which we use to programmatically query park polygons from OpenStreetMap. Blumenberg & King (2024) connected Bay Area housing costs to long commutes, motivating affordability as the central user interaction rather than raw price browsing. Monkkonen et al. (2024) showed that supply-side restrictions inflate California prices 15–30%, reinforcing that buyers need realistic affordability constraints, not just model outputs. Chen & Zheng (2025) demonstrated geographic price visualization at the zip level without interactivity; our dashboard adds real-time filter interaction on top of that foundation.

4. Proposed Method

4.1 Intuition - Why This Should Outperform Existing Tools

Zillow and Redfin are built for browsing homes rather than deciding on one. Their price estimates are proprietary, reflecting the seller's market rather than the buyer's bids. Moreover, they include descriptive neighborhood data which isn't filterable, with their affordability tools entirely disconnected from the map. A buyer who enters their income into a Zillow mortgage calculator receives a loan estimate for a specific property; they do not receive a map of properties that fall within financial limits that they specify, considering the neighborhoods they're interested in, and taking into account the walkability or family suitability of those areas. No existing consumer platform exposes neighborhood quality as a quantifiable, filterable signal at the zip-code level.

Our approach is better for three concrete reasons. First, we built neighborhood quality into the prediction model utilizing a Kid-Friendly Score and Cool Score. This makes our approach particularly well-suited for families and individuals seeking homes in neighborhoods that balance kid-friendliness with vibrant, desirable amenities. Second, the affordability filter operates on predicted prices from our own model, so the buyer's budget constraint and the price estimate share the same underlying logic. Third, everything is unified under one roof complete with everything a potential bay area buyer would need without having to switch tabs.

4.2 Data Collection and Feature Engineering

We assembled property records across all zip codes in Santa Clara, San Mateo, and Alameda counties, producing a dataset of 49,907 transactions spanning 2021–2026. Raw features cover structural attributes (bedrooms, bathrooms, sqft, lot size, year built, stories, parking spaces, pool, fireplace, HOA fee), location identifiers (county, city, zip code, latitude, longitude), and market context (monthly 30-year fixed mortgage rate). From these we derived eleven engineered features: property age (2026 minus year built), a binary new-construction flag (age ≤ 5), bath-to-bed ratio, sqft per bedroom, a composite amenity score (pool + fireplace + parking), an HOA presence flag, and sale year, month, and quarter extracted from the transaction date. Critically, we also computed zip-level and city-level mean sale prices from the training partition and joined them as numeric features, using a form of target encoding that allows the model to learn location price premiums without memorizing individual zip codes.

Two novel neighborhood features were merged at the zip-code level. The Kid-Friendly Score was computed by querying the number of parks per zip code or city from OpenStreetMap via OSMnx and then constructing a weighted composite: 50% raw park count (MinMax-scaled within zip), 30% county-relative percentile rank, and 20% Bay Area-wide percentile rank. The Cool Score was built from Walk Score (30%), restaurant density from TripAdvisor (25%), local attractions (20%), Bike Score (15%), and Transit Score (10%), normalized using the same three-component approach. Transit Score receives the lowest weight because the Bay Area outside San Francisco and Oakland is car-dependent; Walk Score receives the highest because it proxies for the density of walkable amenities that drive day-to-day livability and, as Crompton (2001) shows, property premiums.

Categorical variables (property type, county, city) were one-hot encoded. After encoding and dropping leaky or redundant columns (raw zip code, year built, price-per-sqft, price-per-sqft_log, n_parks, attractions, restaurants, things_to_do_score, etc), the final feature matrix contains 29 columns.

4.3 Regression Models

We trained five models on an 80/20 random split (random seed 42). Linear Regression and Ridge Regression (α = 1.0) operate on StandardScaler-normalized inputs. Random Forest uses 100 estimators, a max depth of 20, and all available CPU cores. Gradient Boosting uses 200 estimators, a learning rate of 0.05, and a max depth of 5. Gradient Boosting uses 100 estimators, a learning rate of 0.1, and a max depth of 6. All tree-based models receive the raw (unscaled) feature matrix; only linear models were scaled.

We selected Gradient Boosting as the production model since it had good R^2 and less error (MAPE) (appendix table 1). The top features by Gradient Boosting with importance are: (1) sqft (77.1%), (2) sale year (5.95%), (3) lot size (5.53%), property type (3.75%) etc. Square footage accounts for over three-quarters of total predictive power, far ahead of all other features (appendix table 2). Lot size also serves as an implicit proxy for property type. Square footage is the most important feature for predicting prices. Lot size & property type also ranks in the top 10 features, serving as an implicit property-type signal. Since condos carry near-zero lot area while single-family homes average 5,000+ sqft, the model learns type-driven pricing without depending entirely on the categorical label. All remaining features (city dummies, amenity scores, HOA, bath/bed ratio) each contribute close to or less than 0.1% individually.

4.4 Address-Level Prediction Function

The predict_price() function takes a street address plus property details, geocodes the address through name with a 1-second rate limiter and an LRU cache (max size=256) to avoid repeated API calls. It engineers all features identically to training, including target-encoded zip and city average sale prices computed from the full dataset. The function then aligns the resulting row to the training schema with missing dummy columns filled as ‘False’ and returns the best-performing model's predicted price alongside a $/sqft figure and a comparison to zip and city averages. Three test cases against publicly known Bay Area addresses returned predictions within 10% of current Zillow estimates.

4.5 Affordability Calculator

The affordability module applies the 28% DTI rule, where the maximum monthly housing cost equals the monthly income multiplied by 0.28. 75% of the result gets allocated to the mortgage principal and interest, and the remaining 25% is reserved for taxes, insurance, and HOA. Given a user-configurable 30-year fixed rate (default 7.0%), the module calculates a user's maximum loan amount and adds the down payment to arrive at a maximum affordability price.

4.6 Interactive Dashboard

This dashboard is an interactive Bay Area housing explorer built as a single-page web app. It loads roughly 50,000 property records and combines Mapbox GL maps with D3 charts and controls. Users can filter by county, property type, and bedrooms, and use two vertical sliders to set a minimum coolness score and a kid-friendly score. The main map encodes predicted sale price (and related home attributes) in circle color and size; hovering shows a popup with key fields. A linked bar chart for average sale price by city updates whenever filters change, so users can see how the filtered subset compares across cities.

A second part of the dashboard is an affordability calculator, users enter gross monthly income, mortgage rate, down payment share, and housing budget share of income; the app estimates monthly budget, max affordable price, down payment, loan amount, and monthly principal and interest. A second map highlights properties that pass the current map filters and fall under that max price, and a sortable-style table (top affordable listings) shows city, county, type, beds, baths, square footage, lot size, predicted sale price, coolness score, and kids score.

5. Evaluation - 5.1 Testbed and Research Questions

All experiments ran on a single workstation with the full around 50,000-property dataset split 80/20 (seed 42). The five models were trained independently with no hyperparameter search beyond the configurations described in Section 4.3. Evaluation metrics are MAPE (primary), R^2, RMSE, and MAE.

Our experiments were designed to answer four questions:
Q1. Which model best predicts Bay Area sale prices, and how much does it outperform a linear baseline?
Q2. Where in the price distribution does the best model succeed and fail?
Q3. Are the Kid-Friendly and Cool Scores materially useful for families looking for areas that are kid friendly and cool with fun things to do?
Q4. Is the best model biased, does it systematically over- or under-predict for counties or price ranges?

5.2 Experiments and Observations

Experiment 1 — Model performance and accuracy (Q1). All five models were trained on 39,928 properties and evaluated on a held-out test set of 9,982 properties. Gradient Boosting achieved the best overall and similar performance. Gradient Boosting had R^2 = 0.9274 and MAPE = 11.33%, and XGBoost performed nearly identically at R^2 = 0.9267 and MAPE = 11.36%, maintaining a difference of less than 0.1 MAPE points. Random Forest follows at MAPE = 11.64%, while both linear models lag considerably at MAPE = 15.88%. All five models exceed R^2 = 0.89, confirming that even linear approaches capture substantial price variance in this dataset. Ensemble methods outperform the linear baseline by approximately 4.5 MAPE points. We select Gradient Boosting as our deployment model given its near-identical accuracy to XGBoost and slightly better performance.

Experiment 2 — Accuracy by price range (Q2). We grouped all 9,982 test properties into six price brackets and measured MAE, MAPE, and within 10% accuracy per bin (Table 3). We noticed that prediction accuracy improves for higher-priced homes in terms of percentage. The $2M–$3M segment achieves the lowest MAPE (10.6%), with 49.4% of predictions within 10% of actual price, followed closely by $3M+ (10.7%) and $1M–$1.5M (11.0%). This pattern suggests that the model is better calibrated for higher-value properties where square footage, lot size, and county are more consistently differentiated. The weakest performance is in the sub-$500K bracket (MAPE = 13.7%, Within 10% = 37.5%), which is the smallest bin containing only 355 test samples. These properties are underrepresented in the Bay Area market and often include atypical asset types whose prices are not well-explained by standard features. While absolute MAE increases with price, rising from $57,458 below $500K to $376,049 above $3M, this pattern reflects differences in price scale rather than model degradation. Overall, 46.4% of predictions fall within 10% of the true price and 87.4% fall within 20%.

Experiment 3 — Bias analysis (Q4). Mean prediction error is +$958 and median error is +$1,825, confirming the model carries only a marginal over-prediction tendency and is essentially unbiased at the center of the distribution. Over-prediction occurs in 50.4% of cases versus under-prediction in 49.6%, indicating near-perfect symmetry. County-level analysis across the three Bay Area counties reveals that San Mateo achieves the lowest MAPE at 11.1% (Within 10% = 47.6%), while Alameda and Santa Clara are tied at 11.4% with within-10% accuracy of 45.7% and 46.5% respectively. The top 10 worst predictions are all over-predictions ranging from +31.6% to +34.3%, concentrated entirely in the sub-$500K and $500K–$1M brackets. This indicates the model systematically over-estimates value for atypical lower-priced properties, likely because the training distribution is dominated by $1M+ Bay Area homes which would cause the model to anchor predictions upward for outlier low-cost properties.

Experiment 4 — Accuracy bands. Across all 9,982 test properties: 23.0% of predictions fall within 5% of true price, 46.4% within 10%, 69.8% within 15%, and 87.4% within 20%. The median absolute percentage error is 10.79%, meaning more than half of all predictions are within approximately 11% of true price. The mean prediction bias is +1.9%, indicating a slight but consistent over-prediction tendency that is present across all price ranges.

Experiment 5 — Address-level prediction validation. Three Bay Area addresses with known property details were passed through the predict_price() function to validate that the feature engineering pipeline generalizes new inputs correctly. Location signal is captured via city and zip code average price lookups derived from the training distribution rather than raw coordinates. The function reconstructs all 15 engineered features (i.e. property age, bath-to-bed ratio, amenity score, and HOA flag) and aligns the resulting row to the full training schema before predicting. All three test cases returned predictions consistent with the model's overall 11.3% MAPE, confirming the inference pipeline correctly mirrors training-time feature construction for unseen addresses.

Overall, each experiment intended to address one of the four questions we sought to answer. We took a qualitative approach for Q3, realizing that beneficial features for the user can mainly be identified by user testing. Q3 is addressed via our interactive dashboard, which visualizes kid-friendly and "cool" scores on a geospatial map to help families identify relevant locations. Since the members of our group are actively part of the Bay Area's home-searching demographic, we validated that these are important attributes we review when making housing decisions.

6. Conclusions and Discussion

This project set out to build something buyers genuinely need: a single place where price prediction, affordability constraints, and neighborhood quality come together. We believe we achieved that goal. Gradient Boost trained on 49,910 Bay Area transactions achieves an R^2 of 0.927 and a MAPE of 11.4 percent, outperforming linear baselines by 4.5 MAPE points. The two novel neighborhood metrics, the Kid-Friendly Score and Cool Score, did not rank among the top 15 predictors in feature importance. This suggests their value lies more in interpretability and buyer decision support than in directly driving price predictions. That distinction does not reduce their importance to the interface, but it does motivate a formal ablation study in future work. The deployed dashboard brings price prediction, affordability filtering, and neighborhood comparison together in one interactive experience.

The dataset comprises real transactions scraped from Redfin and Zillow, grounding all model evaluations in authentic market activity. Despite this, several limitations constrain the current system. First, listing photographs are absent from the feature set entirely. Visual signals such as curb appeal, interior finish quality, natural light, and kitchen renovation level are well documented drivers of home value that cannot be fully captured by tabular features. Incorporating image embeddings via a pre-trained vision model such as ResNet or CLIP would likely reduce MAPE meaningfully, particularly in the $1.5M+ segment where aesthetic premiums are largest. Second, several structurally relevant features were identified but not incorporated within the project timeline, including school district ratings, listing days-on-market, walk score at the parcel level, and historical price appreciation by zip code. These omissions are reflected in the feature importance results, where square footage alone accounts for 77 percent of the model's predictive weight. This concentration would likely distribute more evenly if richer neighborhood-level signals were incorporated. In addition, more expressive model architectures such as TabNet, LightGBM with deeper feature interactions, or a hybrid tabular-image deep learning approach were not explored due to time constraints. Given that the top three features explain over 90% of model variance, a deep learning approach may yield diminishing returns on tabular data alone but would become significantly more valuable once listing images and time-series price signals are integrated.

Looking forward, the most impactful extensions would be integrating listing-level days-on-market data to flag potentially mispriced properties and adding a time-series component to allow buyers to see predicted price trajectories for specific zip codes rather than a single point estimate. Finally, a formal user study comparing decision time and satisfaction between our tool and the current Zillow/Redfin workflow would quantify the practical value of the unified interface.

In terms of effort distribution, all team members contributed a similar amount of effort overall. Linlin led data collection, OSMnx park querying, Kid-Friendly and Cool Score engineering, and geospatial visualization. Linlin and Krishna jointly led ML model training and pipeline development. Krishna led D3.js dashboard development and Mapbox integration. Cody and Divya led model evaluation, ablation experiments, validation analysis, and report writing.

Appendix

Table 1. Model comparison — 20% held-out test set (10,000 properties).

Model	MAE	RMSE	R^2	MAPE
Gradient Boosting	$182,706	$239,313	0.9274	11.33%
XGBoost	$183,361	$240,511	0.9267	11.36%
Random Forest	$187,626	$248,334	0.9218	11.64%
Ridge Regression	$217,011	$282,802	0.8986	15.88%
Linear Regression	$217,013	$282,803	0.8986	15.88%

Table 2. Feature importance for predicting bay area real estate sale price

Rank	Feature	Importance Score	% of Total
1	Square Footage	0.7713	77.13%
2	Sale Year	0.0595	5.95%
3	Lot Size	0.0554	5.54%
4	Property Type — Single Family	0.0375	3.75%
5	County — Alameda	0.0175	1.75%
6	Avg. Zip Code Sale Price (X)	0.0103	1.03%
7	Avg. City Sale Price (Y)	0.0099	0.99%
8	Avg. City Sale Price (X)	0.0098	0.98%
9	Mortgage Rate — 30yr Fixed	0.0093	0.93%
10	Avg. Zip Code Sale Price (Y)	0.0067	0.67%
11	County — San Mateo	0.0063	0.63%
12	HOA Fee	0.0013	0.13%
13	County — Santa Clara	0.0011	0.11%
14	Sale Month	0.0010	0.10%
15	Sqft per Bedroom	0.0007	0.07%

Table 3. Gradient Boosting MAPE by price range

Price Range	Count	MAE	MAPE	Within 10%
Under $500K	355	$57,458	13.7%	37.5%
$500K – $1M	2,495	$92,100	12.0%	44.5%
$1M – $1.5M	2,371	$135,478	11.0%	48.8%
$1.5M – $2M	1,547	$203,681	11.7%	44.8%

$2M – $3M	2,314	$258,786	10.6%	49.4%
$3M+	900	$376,049	10.7%	44.7%

Table 4. Affordability Scenario at 7% interest rate

Buyer Profile	Annual Income	Down Payment	Max Affordable Price	Affordable Properties	% of Market
Entry-Level	$150,000	$100,000	$494,557	1,680	3.4%
Mid-Level	$200,000	$150,000	$676,076	5,455	10.9%
High-Income	$300,000	$300,000	$1,089,114	16,715	33.5%

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